PE Definition Document

Description: A highly efficient, instruction-set driven Processing Element (PE) for accelerating quantized neural network inference.

PE Operation

The PE operates as an instruction-driven hardware accelerator designed to offload computationally intensive neural network inference tasks. Controlled by a Chip Select signal, the PE transitions between a no-operation state and an active execution mode. The PE operates on quantized signed integers, which serve as efficient approximations of the original floating-point number model parameters. When enabled and valid, the internal controller samples the operation code and the inputs data busses, causing activity in the following clock cycle.

Following the input validity signal, two registers ingest signed 32-bit packed integer operands, typically representing neural network inputs and weights. The PE can dynamically reconfigure its arithmetic core to support high-fidelity INT8 operations, high-throughput INT4 operations, or a full 32x32 multi-cycle scalar multiplication.

The computational datapath executes operations sequentially. It first performs a series of Multiply and Accumulate (MAC) operations to collect partial sums, followed by a full 32x32 multiplication to finalize the arithmetic phase into the 64-bit register. Once all math operations are complete, a dedicated Requantizer module shifts the massive 64-bit signed accumulator value down to an 8-bit signed integer. A hardware-based activation function then processes this scaled value to finalize the inference result. This final activation output is driven to the output bus, which is synchronized with an output_valid signal to ensure seamless and accurate integration with the host system.

Power-Up State: The power-up state of the PE is undefined; For a cleared state at power-up, the user must activate the System Reset signal.

System Reset: A synchronous, active-low System Reset input signal clears all internal registers and initializes the datapath to a default idle state with a zeroed accumulator.

ISA (Instruction Set Architecture)

Op-Code Table

Op-Code	Mnemonic	Full Name	Description
00000	NOP	No Operation	The controller takes no action. All datapath modules hold their current state and wait.
00001	RST_ACC	Reset Accumulator	Clears the 64-bit signed accumulator register to zero.
00010	MAC4	Multiply and Accumulate in INT4	PE treats the 32-bit inputs as eight 4-bit signed integers.
00011	MAC8	Multiply and Accumulate in INT8	PE treats the 32-bit inputs as four 8-bit signed integers.
00100	ADD_BIAS	Add Bias	Adds the 32-bit bias to the accumulator from the operand_B bus.
00101	MULT32	32x32 Multiplication	Triggers a 4-cycle multiplication.
00110	SHIFT_RIGHT	Shift Right (Requantize)	Requantizes the 64-bit accumulator down to an 8-bit value. The 6-bit shift amount is read directly from operand_A[5:0].
00111	APPLY_ACT	Apply Activation	Applies an activation function to the scaled 8-bit value.
01000- 11111	RSV	Reserved	Reserved for future instruction expansion.

Table 1. Op-Codes

Instructions Breakdown

- **RST_ACC:** This instruction clears the accumulator. No operands are required on the data buses for the execution of this instruction.
- **MAC8 vs MAC4 Operation:** The MAC instructions perform a parallel dot-product on the operand_A and operand_B inputs. The MAC8 instruction configures the datapath for INT8 precision. The 32-bit buses are interpreted as four packed, signed 8-bit integers, which are multiplied in parallel, then added into a 64-bit accumulator. The MAC4 instruction configures the datapath for INT4 precision. The 32-bit buses are interpreted as eight packed, signed 4-bit integers, which are multiplied in parallel as well, then added to the accumulator.
- **MULT32:** This instruction initiates a full 32x32 signed multiplication over four clock cycles. The hardware calculates the result by multiplying the full 32-bit operand_A by one byte of operand_B per cycle. The PE Controller automatically manages the system stall during this multi-cycle operation.
- **ADD_BIAS:** Bypasses the multiplier array completely. The 32-bit value on the operand_B bus is sign-extended and added directly to the current 64-bit accumulator sum.
- **SHIFT_RIGHT:** The controller pulses the Requantizer module to scale the 64-bit accumulator value down to an 8-bit integer. During the exact clock cycle that this opcode is issued, the lower 6-bits of operand_A 32-bit bus (operand_A[5:0]) serve as a 6-bit shift amount. The hardware uses this immediate value to perform the arithmetic right-shift.
- **APPLY_ACT:** Applying an activation function to the activation input. The unit maps the rescaled 8-bit data into a final non-linear output value.

I/O Interface

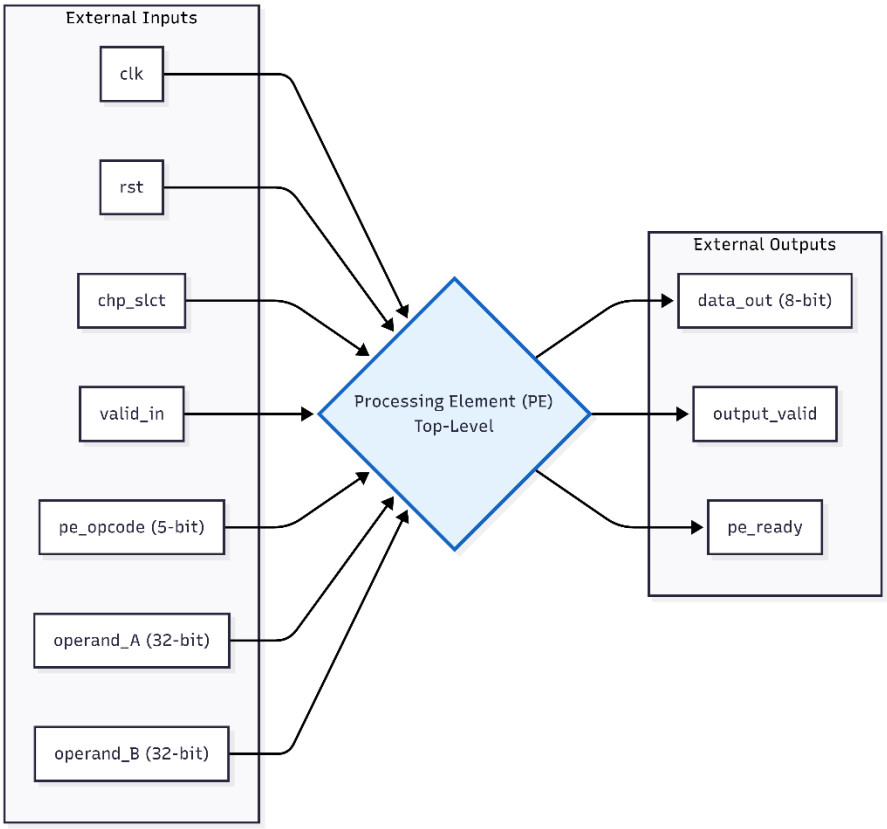

Figure 1. PE Top-Level Interface

The following table defines the top-level input and output ports of the Processing Element. All signals are sampled on the rising edge of the main system clock.

Signal Name	Signal Type	Width	Direction	Function	User Note
clk	Global	1 bit	Input	System Clock	Main system clock.
rst	Global	1 bit	Input	System Reset	Active-low asynchronous reset. Clears all internal registers and initializes the datapath.
chp_slct	Control	1 bit	Input	Chip Select	When inactive, operands and instructions are not sampled. Operations started before inactivity continue until end.
input_valid	Control	1 bit	Input	Input Validity	Signals that the data on the input buses is valid and should be sampled.
pe_opcode	Control	5-bit bus	Input	Operational Code	The global instruction routed to the PE Controller to dictate the current datapath operation.
operand_A	Data	32-bit bus	Input	NN Inputs / Weights / Shift Amount	Packed integer data bus. Contains four signed INT8s, eight INT4s, or a 6-bit shift amount (shamt) during a SHIFT_RIGHT instruction.
operand_B	Data	32-bit bus	Input	NN Inputs / Weights / Bias / Scaling Factor	Packed integer data bus. Contains four signed INT8s, eight INT4s, or a 32-bit scalar/bias.
data_out	Data	8-bit bus	Output	NN Output	The final 8-bit quantized inference result driven by the Activation block.
output_valid	Global	1 bit	Output	Output Validity	Asserted high for one clock cycle following an activation instruction, indicating the data on data_out is a valid, finished result.
pe_ready	Control	1 bit	Output	Ready Flag	Flow-control flag. When high, the PE is idle and ready to accept new instructions. When low, the PE is stalled executing a multi-cycle operation.

Table 2. I/O